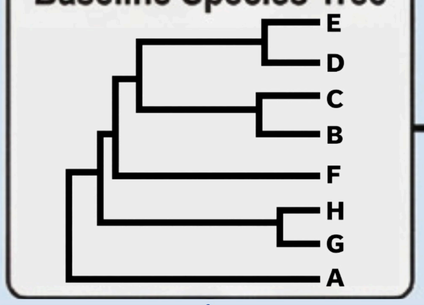
Gene Tree Simulation (SimPhy)

Baseline Species Tree



Four Parameter Classes
(36 Settings, 100 Replicates)

- **Dup/Loss Rate**: 0 (No HP); 3e-4 (Low HP); 4e-4 (High HP)
- **Incomplete Lineage Sorting**: Low ILS; High ILS
- **Individuals / pop**: 1 per taxon; 2 per taxon
- **Rate Variation**: No variation; Gene variation; Lineage variation

Input to Simphy

Initialize iteration i = 1,
K_i = 0 valid gene trees,
N_i = 1000 target gene trees,
M = 20,000 maximum iterations

Simulate N_i gene trees

Remove invalid gene trees
with repeated taxon
with missing taxon

Valid gene trees
N_{valid} out of N_i

Update K
$K_{i+1} = K_i + N_{valid}$

$K_i \geq 1{,}000$ — Yes → **Stop simulation & return** the first min(K,1000) gene trees
... x min(K, 1000)

No

$K_i \geq 800$ — Yes

No

Failed simualtions
Try simulation again with a higher M

Iteration i=i+1
Is i ≤ M? — Yes

No

Update
$N_i = N_{i-1} - K$

Add K_i gene trees into saved gene tree pool
...×K_i

Molecular Sequence Simulation (Seq-Gen)

Simulated Gene Trees
min(1000, k) gene trees per replicate (100 replicates)
min(1000, K)

Seq-Gen parameters

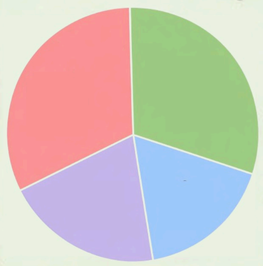
- 1,000 bases per gene
- HKY + Γ model
- gene-specific substitution parameters

multi-sequence alignment (MSA)
per gene (1000 genes per rep)

	Rep 01	Rep 02	
Gene 1	AGGTAG	GGACTA	× 100 reps
Gene 2	AGGGAG	GGACTA	
Gene 3	AAGTAG	GGAGTA	

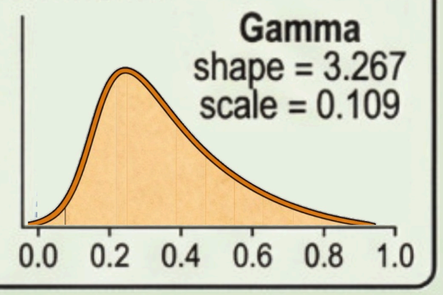
1,000 bases

base frequency

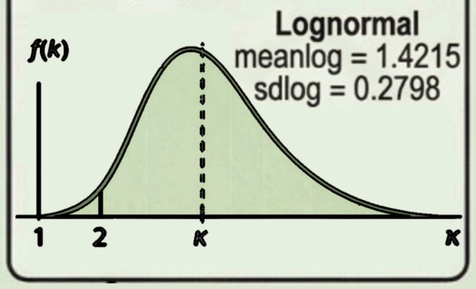


Dirichlet
A = 66.59
C = 38.41
G = 38.61
T = 67.12

among-site rate variation



transition/trasversion Rate



MSA for each rep — x 100 reps

Concatenated MSA per rep (100 rep per setting)

Rep 01	AGGTAG	AGGTAG	AGGTAG ...
Rep 02	GGACTA	GGACTA	GGACTA
...	Gene 1	Gene 2	Gene 3

IQ-TREE

Estimated gene trees

- Rep 01 ...
- Rep 02 ...
- Rep 03 ...
- ... x 100 replicates
- Rep 100 ...

estimated gene trees

ASTRAL

Estimated species trees

- Rep 01
- Rep 02
- Rep 03
- ... x 100 replicates
- Rep 100

estimated species trees